Models and supported functions

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You can train several types of models in IBM® Maximo® Visual Inspection.

Table 1 shows the training models that Maximo Visual Inspection supports and the types of training each one does.

Table 1. Model types

Model type	Description	Image classification	Object detection	Action detection
GoogLeNet	System default when training for image classification.	Yes	No	No
Faster R-CNN	Optimized for accuracy. System default when training for object detection.	No	Yes	No
YOLO v3	Optimized for speed. These models use *you only look once* (YOLO) v3 and take longer to train than other models in this product. Generates a Core ML file.	No	Yes	No
Tiny YOLO v3	Uses the same technique as YOLO v3. Models are faster, less accurate, and use less GPU memory. Alternatively, these models can be deployed on the CPU. Generates a Core ML file.	No	Yes	No
Detectron2	Detectron2 Mask R-CNN models can use objects that are labeled with polygons for greater training accuracy. You can disable segmentation for a shorter training time.	No	Yes	No
High resolution	Optimized for accuracy. Suitable for training and inference on high-resolution images. Supports segmentation to allow small objects, such as fine cracks, to be detected.	No	Yes	No
Single shot detector (SSD)	Used for real-time inference and embedded devices. It is almost as fast as YOLO but not as accurate as Faster R-CNN.	No	Yes	No
Structured segment network (SSN)	Used for video action detection.	No	No	Yes
Anomaly optimized	Optimized to identify anomalous objects.	No	Yes	No
Custom model	Imported model used for training.	Yes	Yes	No

Note: Starting in Maximo Visual Inspection 8.7, custom models are not supported. Custom models are still supported in Maximo Visual Inspection 8.6 and earlier versions.

Note: Starting in Maximo Visual Inspection 9.1, Single Shot Detector (SSD) models are no longer supported for model training. However, you can continue to import and deploy SSD models in Maximo Visual Inspection and Maximo Visual Inspection Edge.

Table 2 shows the functions that are supported for each model type.

Table 2. Supported functions

Model type	Deploy multiple models to one GPU	Deploy to TensorRT	Enable Core ML	Import and export model	Supported on IBM Maximo Visual Inspection Edge	Use for transfer learning	TensorFlow Lite
GoogLeNet	Yes	Yes Version 8.8+ only	Yes	Yes	Yes	Yes	No
Faster R-CNN	Yes	Yes	No	Yes	Yes	Yes	No
YOLO v3	Yes	Yes Version 8.8+ only	Yes	Yes	Yes	Yes	Yes
Tiny YOLO v3	Yes	Yes Version 8.8+ only	Yes	Yes	Yes	Yes Version 8.9+ only	Yes
Detectron2	Yes	No	No	Yes	Yes	Yes	No
High resolution	Yes	No	No	Yes	Yes	Yes	No
Single shot detector (SSD)	Yes	Yes	No	Yes	Yes	Yes	No
Structured segment network (SSN)	No	No	No	Yes	No	No	No
Anomaly optimized	Yes	No	No	Yes	Yes	No	No

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Parent topic:

→ IBM Maximo Visual Inspection overview